

TITLE: CLEANING & ANALYSIS OF RETAIL DATASET.
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COURSE; DATA ANALYSIS/BUSINESS INTELLIGENCE.
DATE; MAY 2026.

INTRODUCTION

This project focuses on cleaning and transforming a retail sales dataset using Microsoft Excel Power Query. The dataset represents sales transactions collected from different store branches across Nigeria.

Since the data was manually entered from multiple sources, it contained several inconsistencies and quality issues such as duplicate records, missing values, inconsistent text formatting, incorrect spellings, mixed data types, and irregular date formats.

The purpose of this project was to practice important data cleaning and transformation techniques in Power Query in order to prepare the dataset for analysis and reporting. Various Power Query features such as removing duplicates, replacing values, splitting and merging columns, changing data types, creating custom and conditional columns, appending queries, and merging queries were applied during the cleaning process.

After cleaning and transforming the dataset, the data became more structured, consistent, and suitable for business analysis and decision-making.

BREID EXPLANATION OF THE DATASET.

The dataset used for this project is a dummy retail sales dataset containing transactional information from multiple retail branches. The dataset includes customer details, product information, sales values, payment methods, and branch locations.

The main dataset served as the Fact Table, while additional dimension tables were created to support query merging and relational modeling.

DATA CLEANING PROCESS.

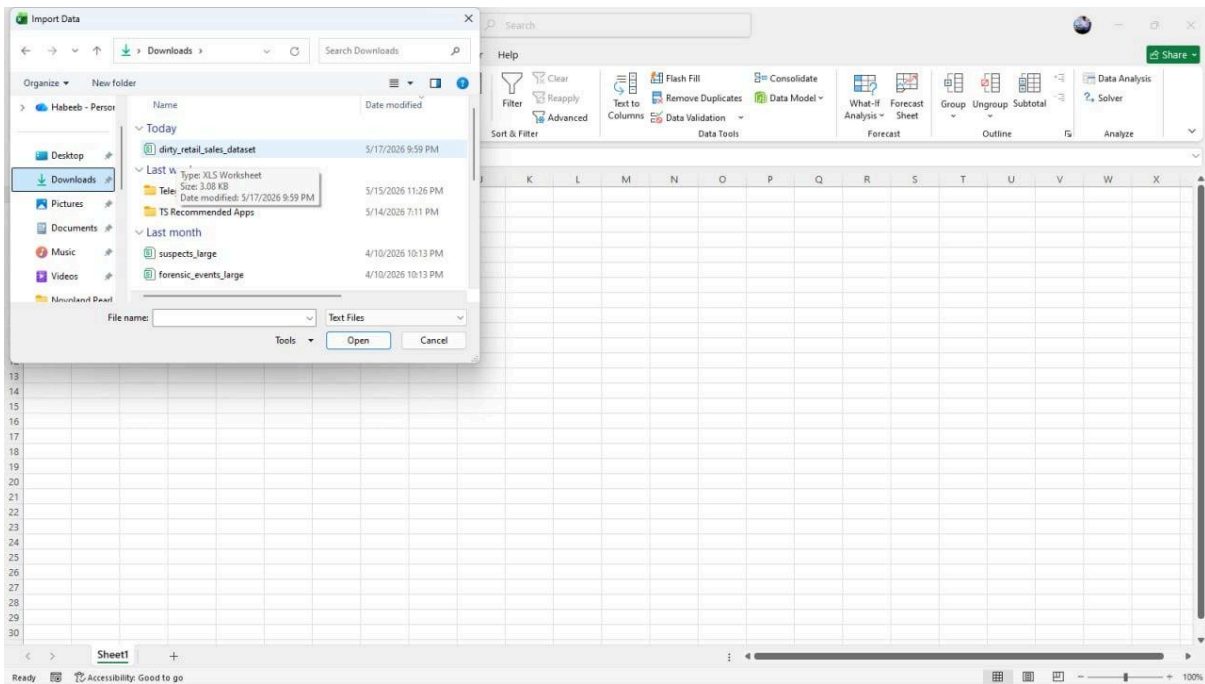
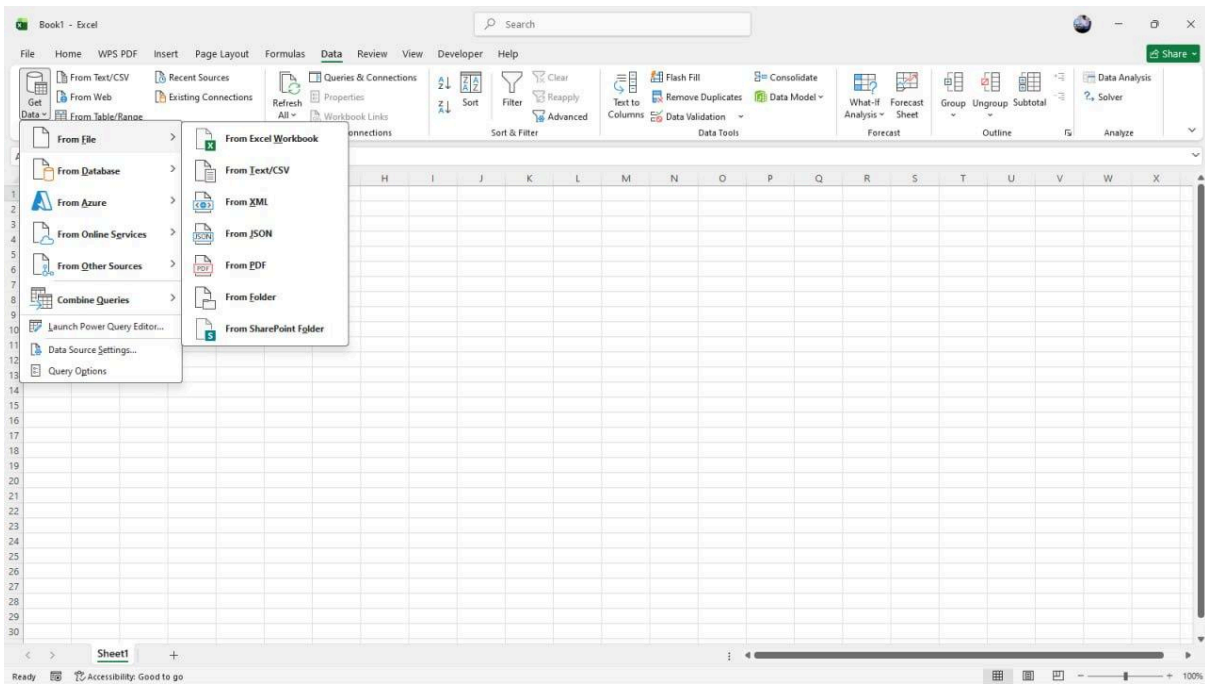
This dataset was cleaned with a technique, column by column. That is, I cleaned the dataset, by making sure one column is neat, consistent, before moving to the other columns. I did this, till the whole columns were cleaned. And here's how I approached it. So first, I'll explain how I imported the data, before I go into the data cleaning process.

DATA IMPORTATION.

From Excel, here is the Data menu >> Get Data ribbon >> From File >> From CSV.

After this, it took me to my local browser and I selected the file.

After selecting the file, it takes me to a page where I have an overview about my data then I had the option to 'Load' or 'Transform Data' and I pressed 'Transform Data' to load it into power query.



dirty_retail_sales_dataset.csv

File Origin: 65001:Unicode (UTF-8) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Order_ID	Customer_Name	Gender	Branch	Product	Category	Quantity	Unit_Price	Total_Sales	Order_Date	Payment_Method	Customer_Email	Customer_Location	Discount_Code	Staff_Notes
ORD001	John Doe	male	Lagos Branch	Laptop	Electronics	2	N150000	700000	12/01/2024	Card	john.doe@gmail.com	Lagos-Nigeria	DISC10	urgent delivery
ORD002	Mary Ann	F	Abuja Branch	Phone	Electronics	three	N150000	450000	2024-01-15	Cash	maryann@yahoo.com	Abuja-Nigeria		late payment
ORD003	Peter Obi	Male	Kano Branch	laptop	Electronics	1	400000NGN	400000	15-01-2024	Transfer	peterobi@gmail	Kano-Nigeria	DISC5	VIP customer
ORD003	Peter Obi	Male	Kano Branch	laptop	Electronics	1	400000NGN	400000	15-01-2024	Transfer	peterobi@gmail	Kano-Nigeria	DISC5	VIP customer
ORD004	Chioma James	female	LAGOS branch	Printer	Office Equipment	4	N85000	340000	01/18/2024	card	chioma@gmail.com	Lagos-Nigeria		needs follow-up
ORD005	David Musa	M	Abuja Branch	Desk		2	120000NGN	240000	2024/01/20	CASH	devidmusa@gmail.com	Abuja-Nigeria	DISC20	
ORD006	Esther Ali	Female	Kaduna Branch	Phone	Electronics	five	N160000	800000	21-01-2024	Transfer	estherali@gmail.com	Kaduna-Nigeria		customer complained
ORD007	Samuel Ade	male	Port Harcourt Branch	LAPTOP	electronics	1	N355000	355000	2024-01-22	Card	samuelade@gmail.com	Port Harcourt-Nigeria	DISC10	deliver ASAP
ORD008	Nguzi Eze	FEMALE	Lagos Branch	Chair	Furniture	6	N25000	150000	23/01/2024		nguzi@gmail.com	Lagos-Nigeria		damaged package
ORD009	Ibrahim Bello	M	Kano branch	Table	Furniture	2	N45000	90000	24-01-2024	Cash	ibrahimbello@gmail.com	Kano-Nigeria	DISC5	
ORD010	Ada White	Female	Ibadan Branch	printer	Office equipment	1	N80000	80000	2024.01.25	Card	adawhite@gmail	Ibadan-Nigeria		repeat customer
ORD011	John Doe	male	Lagos Branch	Laptop	Electronics	2	N350000	700000	12/01/2024	Card	john.doe@gmail.com	Lagos-Nigeria	DISC10	urgent delivery
ORD012	Kelvin Hart	Male	Abuja Branch	Phone	Electronics	3	N155000	465000	26/01/2024	Transfer	kelvinhart@gmail.com	Abuja-Nigeria		new customer
ORD013	Blessing Sam	F	Lagos Branch	Monitor	Electronics	2	N95000	190000	27-01-2024	Cash	blessingsam@gmail.com	Lagos-Nigeria		delayed response
ORD014	Michael King	male	Kaduna branch	Desk	Furniture	one	N125000	125000	2024/01/28	Card	michaelking@gmail.com	Kaduna-Nigeria	DISC15	
ORD015	Rita Daniels	Female	Port Harcourt Branch	Phone	Electronics	4	N158000	632000	29/01/2024	transfer	ritadaniels@gmail.com	Port Harcourt-Nigeria		priority customer
ORD016	Grace Peter	Female	Lagos Branch	Tablet	Electronics	2	N120000	240000	30-01-2024	Card	gracepeter@gmail.com	Lagos-Nigeria		pending confirmation
ORD017	Musa Ibrahim	M	Abuja Branch	Chair	Furniture	7	N27000	189000	31/01/2024	Cash	musaibrahim@gmail.com	Abuja-Nigeria		wrong address
ORD018	Janet Cole	female	Kano Branch	Phone	Electronics	2	N150000	300000	2024-02-01	Transfer	janetcole@gmail.com	Kano-Nigeria	DISC5	
ORD019	Paul Adams	Male	Lagos Branch	Laptop	Electronics	1	N360000	360000	02/02/2024	Card	pauladams@gmail.com	Lagos-Nigeria		high priority

The data in the preview has been truncated due to size limits.

Load Transform Data Cancel

dirty_retail_sales_dataset - Power Query Editor

Table.TransformColumnTypes(*Promoted Headers,{"Order_ID", type text}, {"Customer_Name", type text}, {"Gender", type text}, {"Branch", type text}, {"Product", type text}, {"Category", type text}, {"Quantity", type text})

Order_ID	Customer_Name	Gender	Branch	Product	Category	Quantity
ORD0001	John Doe	male	Lagos Branch	Laptop	Electronics	2
ORD0002	Mary Ann	F	Abuja Branch	Phone	Electronics	three
ORD0003	Peter Obi	Male	Kano Branch	laptop	Electronics	1
ORD0003	Peter Obi	Male	Kano Branch	laptop	Electronics	1
ORD0004	Chioma James	female	LAGOS branch	Printer	Office Equipment	4
ORD0005	David Musa	M	Abuja Branch	Desk		2
ORD0006	Esther Ali	Female	Kaduna Branch	Phone	Electronics	five
ORD0007	Samuel Ade	male	Port Harcourt Branch	LAPTOP	electronics	1
ORD0008	Nguzi Eze	FEMALE	Lagos Branch	Chair	Furniture	6
ORD0009	Ibrahim Bello	M	Kano branch	Table	Furniture	2
ORD0010	Ada White	Female	Ibadan Branch	printer	Office equipment	1
ORD0011	John Doe	male	Lagos Branch	Laptop	Electronics	2
ORD0012	Kelvin Hart	Male	Abuja Branch	Phone	Electronics	3
ORD0013	Blessing Sam	F	Lagos Branch	Monitor	Electronics	2
ORD0014	Michael King	male	Kaduna branch	Desk	Furniture	one
ORD0015	Rita Daniels	Female	Port Harcourt Branch	Phone	Electronics	4
ORD0016	Grace Peter	Female	Lagos Branch	Tablet	Electronics	2
ORD0017	Musa Ibrahim	M	Abuja Branch	Chair	Furniture	7
ORD0018	Janet Cole	female	Kano Branch	Phone	Electronics	2
ORD0019	Paul Adams	Male	Lagos Branch	Laptop	Electronics	1
ORD0020	Sandra Lee	F	Ibadan Branch	Table	Furniture	3

15 COLUMNS, 21 ROWS | Column profiling based on top 1000 rows | PREVIEW DOWNLOADED AT 10:06 PM

CLEANING COLUMN 1

Problem; On the column profile of column 1, it says there are 20 distinct values and only 19 are unique. Which means, there's a duplicate in the column.

Solution; To handle this, on the Home menu, Remove rows ribbon, I pressed the drop-down and selected 'remove duplicates' then it changed to 20 distinct, 20 unique.

Power Query Editor interface showing the initial data load and transformation steps for the 'dirty_retail_sales_dataset'.

Initial Query: Table.Distinct(#"Changed Type", {"Order_ID"})

Columns: Order_ID, Customer_Name, Gender, Product, Category, Quantity

Applied Steps: Removed Duplicates

Transformation: Table.TransformColumns(#"Removed Duplicates",{"Customer_Name", Text.Trim, type text})

Columns: Order_ID, Customer_Name, Gender, Branch, Product, Category, Quantity

Applied Steps: Trimmed Text

The data table shows 20 rows of sales data with columns: Order_ID, Customer_Name, Gender, Branch, Product, Category, and Quantity. The Gender column contains inconsistent values like 'female', 'Female', 'f', 'F', 'm', 'M', 'Male', and 'male'. The Branch column contains values like 'Lagos Branch', 'Abuja branch', 'Kano Branch', 'LAGOS branch', 'Abuja Branch', 'Kaduna Branch', 'Port Harcourt Branch', 'Lagos Branch', 'Kano branch', 'Ibadan Branch', 'printer', 'Office equipment', 'Laptop', 'Phone', 'Monitor', 'Desk', 'Tablet', 'Chair', 'Furniture', 'electronics', 'three', 'four', 'five', 'one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten', 'eleven', 'twelve', 'thirteen', 'fourteen', 'fifteen', 'sixteen', 'seventeen', 'eighteen', 'nineteen', 'twenty'.

For column GENDER

Problem; The values were so inconsistent. They were female, Female, f, F, m, M, Male, male. So I needed to standardize this, and also ensure consistency.

Solution; First, I decided that I would be going for the char type, that is F or M.

So, I first extracted the first letter in each word. To do this, I went to the Transform menu, Split column drop-down ribbon, then I selected 'split by number of characters'. The dialog box came up, as I entered 1, to signify just one character. And I also identified that it should be from the left. Hence, the data was splitted.

dirty_retail_sales_dataset - Power Query Editor

File Home Transform Add Column View

Group Use First Row as Headers Table

Transpose Reverse Rows Count Rows

Data Type: Text

Replace Values

Unpivot Columns

Move

Convert to List

Any Column

Text Column

Number Column

Date & Time Column

Structured Column

Statistics Standard Scientific Rounding Information

Expand Aggregate Extract Values

Query Settings

Table.TransformColumns({"Removed Duplicates",{"Customer_Name", Text.Trim, type text}})

dirty_retail_sales...

Order_ID Customer_Name Gender Branch Product Category Quantity

Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 95% Valid 100%

Error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error 0%

Empty 0% Empty 0% Empty 0% Empty 0% Empty 5% Empty 0%

20 distinct, 20 unique 19 distinct, 18 unique 4 distinct, 0 unique 11 distinct, 5 unique 11 distinct, 5 unique 6 distinct, 4 unique 9 distinct, 5 unique 18 d

1 OR0001 John Doe male Kaduna Branch Desk Furniture one N121

2 OR0002 mary ann Female Port Harcourt Branch Phone Electronics 4 N151

3 OR0003 Peter Obi Female Lagos Branch Tablet Electronics 2 N121

4 OR0004 Chioma James M Abuja Branch Chair Furniture 7 N271

5 OR0005 David Musa female Kano Branch Phone Electronics 2 N151

6 OR0006 Esther Ali Male Lagos Branch Laptop Electronics 1 N351

7 OR0007 Samuel Ade female Ibadan Branch Table Furniture 3 N501

8 OR0008 Ngozi Eze male

9 OR0009 Ibrahim Bello male

10 OR0010 Ada White female

11 OR0011 John Doe male

12 OR0012 Kelvin Hart female

13 OR0013 Blessing Sam male

14 OR0014 Michael King female

15 OR0015 Rita Daniels male

16 OR0016 Grace Peter female

17 OR0017 Musa Ibrahim M

18 OR0018 Janet Cole female

19 OR0019 Paul Adams male

20 OR0020 Sandra Lee F

Split Column by Number of Characters

Specify the number of characters used to split the text column.

Number of characters

1

Split

Once, as far left as possible

Once, as far right as possible

Repeatedly

Advanced options

OK Cancel

15 COLUMNS, 20 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 10:11 PM

dirty_retail_sales_dataset - Power Query Editor

File Home Transform Add Column View

Group Use First Row as Headers Table

Transpose Reverse Rows Count Rows

Data Type: Text

Replace Values

Unpivot Columns

Move

Convert to List

Any Column

Text Column

Number Column

Date & Time Column

Structured Column

Statistics Standard Scientific Rounding Information

Expand Aggregate Extract Values

Query Settings

Table.RemoveColumns({"Changed Type1",{"Gender.2"}})

dirty_retail_sales...

Order_ID Customer_Name Gender.1 Branch Product Category Quantity

Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 95% Valid 100%

Error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error 0%

Empty 0% Empty 0% Empty 0% Empty 0% Empty 5% Empty 0%

20 distinct, 20 unique 19 distinct, 18 unique 4 distinct, 0 unique 11 distinct, 5 unique 11 distinct, 5 unique 6 distinct, 4 unique 9 distinct, 5 unique 18 d

1 OR0001 John Doe m Lagos Branch Laptop Electronics 2 N351

2 OR0002 mary ann F Abuja Branch Phone Electronics three N151

3 OR0003 Peter Obi M Kano Branch laptop Electronics 1 4000

4 OR0004 Chioma James f LAGOS branch Printer Office Equipment 4 N451

5 OR0005 David Musa M Abuja Branch Desk 2 1200

6 OR0006 Esther Ali F Kaduna Branch Phone Electronics five N161

7 OR0007 Samuel Ade m Port Harcourt Branch LAPTOP electronics 1 N151

8 OR0008 Ngozi Eze F Lagos Branch Chair Furniture 6 N251

9 OR0009 Ibrahim Bello M Kano branch Table Furniture 2 N451

10 OR0010 Ada White F Ibadan Branch printer Office equipment 1 N801

11 OR0011 John Doe m Lagos Branch Laptop Electronics 2 N351

12 OR0012 Kelvin Hart M Abuja Branch Phone Electronics 3 N151

13 OR0013 Blessing Sam F Lagos Branch Monitor Electronics 2 N951

14 OR0014 Michael King m Kaduna branch Desk Furniture one N121

15 OR0015 Rita Daniels F Port Harcourt Branch Phone Electronics 4 N151

16 OR0016 Grace Peter F Lagos Branch Tablet Electronics 2 N121

17 OR0017 Musa Ibrahim M Abuja Branch Chair Furniture 7 N271

18 OR0018 Janet Cole f Kano Branch Phone Electronics 2 N151

19 OR0019 Paul Adams M Lagos branch Laptop Electronics 1 N351

20 OR0020 Sandra Lee F Ibadan Branch Table Furniture 3 N501

15 COLUMNS, 20 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 10:13 PM

After that, it was splitted and then, the consistency was left with capital letter or small letter. So I formatted it to be CAPITAL LETTER. Transform menu, format tab, CAPITALIZE EACH WORD.

Power Query Editor - dirty_retail_sales_dataset

Table.TransformColumns(#"Filtered Rows",{{{"Branch", Text.Proper, type text}}})

Query Settings

Properties

Name: dirty_retail_sales_dataset

Applied Steps

- Source
- Promoted Headers
- Changed Type
- Removed Duplicates
- Trimmed Text
- Split Column by Position
- Changed Type1
- Removed Columns
- Renamed Columns
- Uppercased Text
- Filtered Rows
- Capitalized Each Word

Order_ID	Customer_Name	Gender	Branch	Product	Category	Quantity
1	John Doe	M				
2	mary ann	F				
3	Peter Obi	M				
4	Chioma James	F				
5	David Musa	M				
6	Esther Ali	F				
7	Samuel Ade	M				
8	Ngozi Eze	F				
9	Ibrahim Bello	M				
10	Ada White	F				
11	John Doe	M				
12	Kelvin Hart	M				
13	Blessing Sam	F				
14	Michael King	M				
15	Rita Daniels	F				
16	Grace Peter	F				
17	Musa Ibrahim	M	Abuja Branch	Chair	Furniture	7
18	Janet Cole	F	Kano Branch	Phone	Electronics	2
19	Paul Adams	M	Lagos Branch	Laptop	Electronics	1
20	Sandra Lee	F	Ibadan Branch	Table	Furniture	3

Power Query Editor - dirty_retail_sales_dataset

Table.TransformColumns(#"Filtered Rows",{{{"Branch", Text.Proper, type text}, {"Product", Text.Proper, type text}}})

Query Settings

Properties

Name: dirty_retail_sales_dataset

Applied Steps

- Source
- Promoted Headers
- Changed Type
- Removed Duplicates
- Trimmed Text
- Split Column by Position
- Changed Type1
- Removed Columns
- Renamed Columns
- Uppercased Text
- Filtered Rows
- Capitalized Each Word

Replace Values

Replace one value with another in the selected columns.

Value To Find: Laptop

Replace With: Laptop

Advanced options

Order_ID	Customer_Name	Gender	Branch	Product	Category	Quantity
1	John Doe	M				
2	mary ann	F				
3	Peter Obi	M				
4	Chioma James	F				
5	David Musa	M				
6	Esther Ali	F				
7	Samuel Ade	M				
8	Ngozi Eze	F				
9	Ibrahim Bello	M				
10	Ada White	F				
11	John Doe	M				
12	Kelvin Hart	M				
13	Blessing Sam	F	Lagos Branch	Monitor	Electronics	2
14	Michael King	M	Kaduna Branch	Desk	Furniture	one
15	Rita Daniels	F	Port Harcourt Branch	Phone	Electronics	4
16	Grace Peter	F	Lagos Branch	Tablet	Electronics	2
17	Musa Ibrahim	M	Abuja Branch	Chair	Furniture	7
18	Janet Cole	F	Kano Branch	Phone	Electronics	2
19	Paul Adams	M	Lagos Branch	Laptop	Electronics	1
20	Sandra Lee	F	Ibadan Branch	Table	Furniture	3

The first screenshot shows the Power Query Editor with the following formula bar: `= Table.TransformColumns(#"Filtered Rows",{{"Branch", Text.Proper, type text}, {"Product", Text.Proper, type text}})`. The 'Replace Values' dialog box is open, showing 'Value To Find' as 'Laptop' and 'Replace With' as 'Laptop'. The 'Applied Steps' pane on the right lists the following steps: Source, Promoted Headers, Changed Type, Removed Duplicates, Trimmed Text, Split Column by Position, Changed Type1, Removed Columns, Renamed Columns, Uppercased Text, Filtered Rows, Capitalized Each Word, and Replaced Value.

The second screenshot shows the Power Query Editor with the following formula bar: `= Table.ReplaceValue(#"Capitalized Each Word", "Laptop", "Laptop", Replacer.ReplaceText, {"Product"})`. The 'Replace Values' dialog box is open, showing 'Value To Find' as 'Table' and 'Replace With' as 'Tablet'. The 'Applied Steps' pane on the right lists the following steps: Source, Promoted Headers, Changed Type, Removed Duplicates, Trimmed Text, Split Column by Position, Changed Type1, Removed Columns, Renamed Columns, Uppercased Text, Filtered Rows, Capitalized Each Word, and Replaced Value.

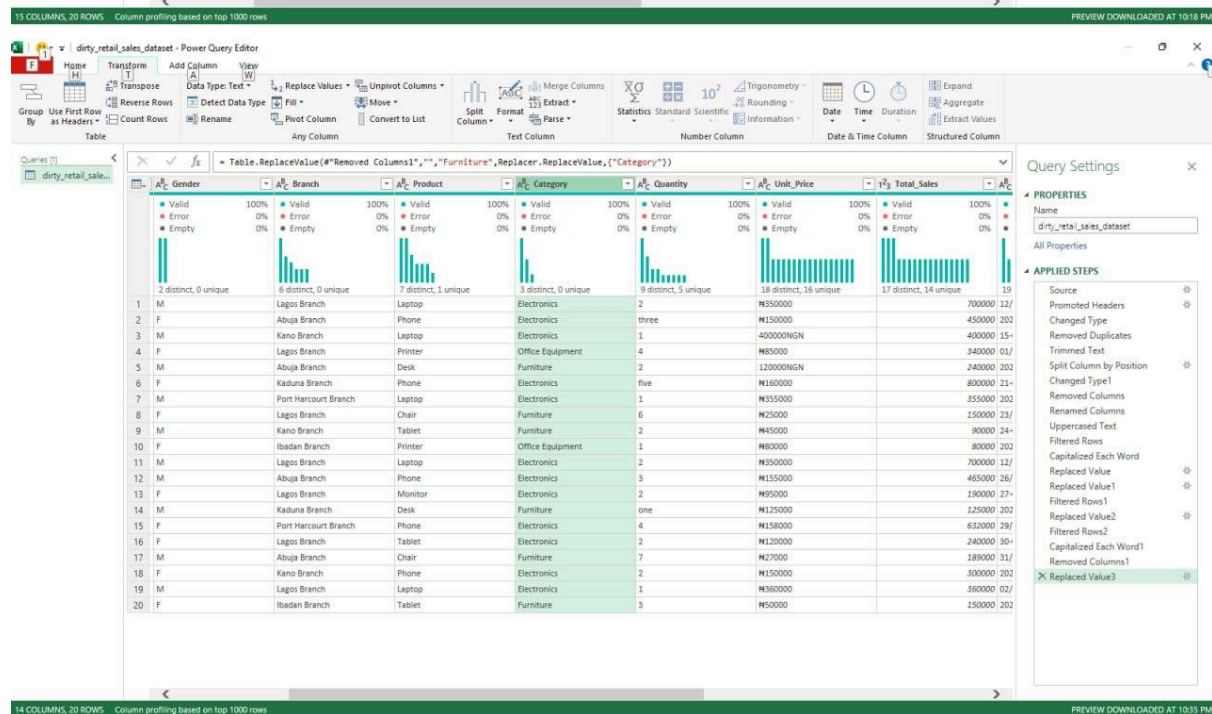
For column CATEGORY.

Problem; There was blanks, and inconsistencies in capitalization.

Solution; For the blank, I checked the corresponding product and found out it was desk, which means the category will be furniture.

So I inputted.

Also, I capitalized each word to solve the inconsistency.



Some values were written in words, so I just did 'replace values' to change it back to number format.

dirty_retail_sales_dataset - Power Query Editor

File Home Transform Add Column View

Transposing, Reversing Rows, Counting Rows, Defecting Data Type, Renaming, Pivoting Column, Unpivot Columns, Move, Convert to List, Merge Columns, Split Column, Format, Extract, Parse, Statistics, Standard Scientific, Trigonometry, Rounding, Information, Date, Time, Duration, Expand, Aggregate, Extract Values, Structured Column

Query Settings

PROPERTIES

Name: dirty_retail_sales_dataset

APPLIED STEPS

1. Replaced Value3

2. Replaced Value2

3. Filtered Rows2

4. Capitalized Each Word1

5. Removed Columns1

6. Replaced Value3

Table: ReplaceValue(#"Removed Columns1", "Furniture", Replacer.ReplaceValue, {"Category"})

Product	Category	Quantity	Unit Price	Total Sales	Order Date	Payment
Laptop	Electronics	2	M35			
Phone	Electronics	three	M15			
Laptop	Electronics	1	4000			
Printer	Office Equipment	4	M85			
Desk	Furniture	2	1200			
Phone	Electronics	five	M16			
Laptop	Electronics	1	M35			
Chair	Furniture	6	M25			
Tablet	Furniture	2	M45			
Printer	Office Equipment	1	M80			
Laptop	Electronics	2	M35			
Phone	Electronics	3	M15			
Monitor	Electronics	2	M95			
Desk	Furniture	one	M125000	125000	2024/01/28	Card
Phone	Electronics	4	M158000	632000	29/01/2024	transfer
Tablet	Electronics	2	M120000	240000	30-01-2024	Card
Chair	Furniture	7	M27000	189000	31/01/2024	Cash
Phone	Electronics	2	M150000	300000	2024-02-01	Transfer
Laptop	Electronics	1	M360000	360000	02/02/2024	Card

dirty_retail_sales_dataset - Power Query Editor

File Home Transform Add Column View

Transposing, Reversing Rows, Counting Rows, Defecting Data Type, Renaming, Pivoting Column, Unpivot Columns, Move, Convert to List, Merge Columns, Split Column, Format, Extract, Parse, Statistics, Standard Scientific, Trigonometry, Rounding, Information, Date, Time, Duration, Expand, Aggregate, Extract Values, Structured Column

Query Settings

PROPERTIES

Name: dirty_retail_sales_dataset

APPLIED STEPS

1. Replaced Value3

2. Replaced Value2

3. Filtered Rows2

4. Capitalized Each Word1

5. Removed Columns1

6. Replaced Value3

7. Replaced Value4

Table: ReplaceValue(#"Replaced Value3", "three", "3", Replacer.ReplaceText, {"Quantity"})

Product	Category	Quantity	Unit Price	Total Sales	Order Date	Payment
Laptop	Electronics	2	M			
Phone	Electronics	3	M			
Laptop	Electronics	1	4			
Printer	Office Equipment	4	M			
Desk	Furniture	2	1			
Phone	Electronics	five	M			
Laptop	Electronics	1	M			
Chair	Furniture	6	M			
Tablet	Furniture	2	M			
Printer	Office Equipment	1	M80000	80000	2024-01-29	Card
Laptop	Electronics	2	M350000	700000	12/01/2024	Card
Phone	Electronics	3	M155000	465000	26/01/2024	Transfer
Monitor	Electronics	2	M95000	190000	27-01-2024	Cash
Desk	Furniture	one	M125000	125000	2024/01/28	Card
Phone	Electronics	4	M158000	632000	29/01/2024	transfer
Tablet	Electronics	2	M120000	240000	30-01-2024	Card
Chair	Furniture	7	M27000	189000	31/01/2024	Cash
Phone	Electronics	2	M150000	300000	2024-02-01	Transfer
Laptop	Electronics	1	M360000	360000	02/02/2024	Card
Tablet	Furniture	3	M50000	150000	2024/02/03	CASH

dirty_retail_sales_dataset - Power Query Editor

File Home Transform Add Column View

Transpose Reverse Rows Count Rows Group Use First Row as Headers Table

Data Type Text Detect Data Type Rename

Replace Values Fill Pivot Column Convert to List

Unpivot Columns Move to List

Split Column Format Extract Parse

Statistics Standard Scientific Information

Trigonometry Rounding

Date Time Duration

Expand Aggregate Extract Values

Query Settings

Table.TransformColumnTypes(#"Replaced Value6",{"Quantity", type number})

Category	Quantity	Unit Price	Total Sales	Order Date	Payment Method	Customer Email	Customer
1 ronics	2	N8350000	700000	12/01/2024	Card	johnndoe@gmail.com	Lagos-Nig
2 ronics	3	N1500000	450000	2024-01-15	Cash	maryann@yahoo.com	Abuja-Nig
3 ronics	1	400000NGN	400000	15-01-2024	Transfer	peterobi@gmail.com	Kano-Nig
4 e Equipment	4	N850000	340000	01/18/2024	card	chioma@gmail.com	Lagos-Nig
5 ture	2	1200000NGN	240000	2024/01/20	CASH	davidmusa@gmail.com	Abuja-Nig
6 ronics	5	N1600000	800000	21-01-2024	Transfer	estherali@gmail.com	Kaduna-N
7 ronics	1	N3550000	355000	2024-01-22	Card	samuelade@gmail.com	Port Haro
8 ture	6	N250000	150000	23/01/2024	Card	ngot@gmail.com	Lagos-Nig
9 ture	2	N450000	90000	24-01-2024	Cash	ibrahimbello@gmail.com	Kano-Nig
10 e Equipment	2	N800000	80000	2024 01 25	Card	adawhite@gmail.com	Ibadan-Ni
11 ronics	2	N3500000	700000	12/01/2024	Card	johnndoe@gmail.com	Lagos-Nig
12 ronics	3	N1550000	465000	26/01/2024	Transfer	kevinhart@gmail.com	Abuja-Nig
13 ronics	2	N950000	190000	27-01-2024	Cash	blissingsam@gmail.com	Lagos-Nig
14 ture	1	N1250000	125000	2024/01/28	Card	michaeking@gmail.com	Kaduna-N
15 ronics	4	N1580000	632000	29/01/2024	transfer	ritadaniel@gmail.com	Port Haro
16 ronics	2	N1200000	240000	30-01-2024	Card	gracepeter@gmail.com	Lagos-Nig
17 ture	7	N270000	189000	31/01/2024	Cash	musabrahim@gmail.com	Abuja-Nig
18 ronics	2	N1500000	300000	2024-02-01	Transfer	janetcole@gmail.com	Kano-Nig
19 ronics	1	N3600000	360000	02/02/2024	Card	pauladams@gmail.com	Lagos-Nig
20 ture	3	N500000	150000	2024/02/03	CASH	sandralee@gmail.com	Ibadan-Ni

14 COLUMNS, 20 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 10:37 PM

For column UNIT PRICE.

Problem; There were some values that did not have # in from and there were some values that did not have NGN at the end.

Solution; I scrapped both # and NGN, by this, I used replace values and I replaced them with blank, to remove the values.

dirty_retail_sales_dataset - Power Query Editor

File Home Transform Add Column View

Transpose Reverse Rows Count Rows Group Use First Row as Headers Table

Data Type Text Detect Data Type Rename

Replace Values Fill Pivot Column Convert to List

Unpivot Columns Move to List

Split Column Format Extract Parse

Statistics Standard Scientific Information

Trigonometry Rounding

Date Time Duration

Expand Aggregate Extract Values

Query Settings

Table.ReplaceValue(#"Replaced Value7", "NGN", "", Replacer.ReplaceText, {"Unit Price"})

Category	Quantity	Unit Price	Total Sales	Order Date	Payment Method	Customer Email	Customer
1 ronics	2	3500000	700000	12/01/2024	Card	johnndoe@gmail.com	Lagos-Nig
2 ronics	3	1500000	450000	2024-01-15	Cash	maryann@yahoo.com	Abuja-Nig
3 ronics	1	400000	400000	15-01-2024	Transfer	peterobi@gmail.com	Kano-Nig
4 e Equipment	4	850000	340000	01/18/2024	card	chioma@gmail.com	Lagos-Nig
5 ture	2	1200000	240000	2024/01/20	CASH	davidmusa@gmail.com	Abuja-Nig
6 ronics	5	1600000	800000	21-01-2024	Transfer	estherali@gmail.com	Kaduna-N
7 ronics	1	3550000	355000	2024-01-22	Card	samuelade@gmail.com	Port Haro
8 ture	6	250000	150000	23/01/2024	Card	ngot@gmail.com	Lagos-Nig
9 ture	2	450000	90000	24-01-2024	Cash	ibrahimbello@gmail.com	Kano-Nig
10 e Equipment	2	800000	80000	2024 01 25	Card	adawhite@gmail.com	Ibadan-Ni
11 ronics	2	3500000	700000	12/01/2024	Card	johnndoe@gmail.com	Lagos-Nig
12 ronics	3	1550000	465000	26/01/2024	Transfer	kevinhart@gmail.com	Abuja-Nig
13 ronics	2	950000	190000	27-01-2024	Cash	blissingsam@gmail.com	Lagos-Nig
14 ture	1	1250000	125000	2024/01/28	Card	michaeking@gmail.com	Kaduna-N
15 ronics	4	1580000	632000	29/01/2024	transfer	ritadaniel@gmail.com	Port Haro
16 ronics	2	1200000	240000	30-01-2024	Card	gracepeter@gmail.com	Lagos-Nig
17 ture	7	270000	189000	31/01/2024	Cash	musabrahim@gmail.com	Abuja-Nig
18 ronics	2	1500000	300000	2024-02-01	Transfer	janetcole@gmail.com	Kano-Nig
19 ronics	1	3600000	360000	02/02/2024	Card	pauladams@gmail.com	Lagos-Nig
20 ture	3	500000	150000	2024/02/03	CASH	sandralee@gmail.com	Ibadan-Ni

14 COLUMNS, 20 ROWS Column profiling based on top 1000 rows

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For column ORDER DATE.

I changed the data type, using Locale, and I chose English, Nigeria.

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Group Use First Row as Headers Count Rows Table

Transpose Reverse Rows Reverse Columns Count Rows Table

Data Type Text Detect Data Type Fill Pivot Column Convert to List Any Column

Replace Values Unpivot Columns Move Split Column Format Extract Parse Text Column

Merge Columns Statistics Standard Scientific Rounding Information Number Column

Trigonometry Date Time Duration Expand Aggregate Extract Values Date & Time Column Structured Column

Queries dirty_retail_sales...

Table.ReplaceValue(#Replaced Value8,"01/18/2024","01/8/2024",Replacer.ReplaceText,{"Order_Date"})

Category 1.2 Quantity Unit_Price Total_Sales Order_Date Payment_Method Customer_Email

valid 100% Valid 100% Valid 100% Valid 1.2 Decimal Number valid 95% Valid 100% Valid
error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error
empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty

distinct, 0 unique 7 distinct, 3 unique 17 distinct, 14 unique 17 distinct, 14 unique 18 distinct, 16 unique 7 distinct, 3 unique 13 distinct, 18 unique 7 distinct

Category	1.2	Quantity	Unit_Price	Total_Sales	Order_Date	Payment_Method	Customer_Email	Customer_Location
1 ironics		2	350000	700000	2024.01.25	Card	johnndoe@gmail.com	Lagos-Nig
2 ironics		3	150000	450000	12/01/2024	Card	maryann@yahoo.com	Abuja-Nig
3 ironics		1	400000	400000	26/01/2024	Transfer	peterobi@gmail.com	Kano-Nig
4 e Equipment		4	85000	340000	27-01-2024	Cash	chioma@gmail.com	Lagos-Nig
5 iture		2	120000	240000	2024/01/28	Card	davidmusa@gmail.com	Abuja-Nig
6 ironics		5	160000	800000	30-01-2024	Transfer	estherali@gmail.com	Kaduna-N
7 ironics		1	355000	355000	29/01/2024	transfer	samuelade@gmail.com	Port Haro
8 iture		6	25000	150000	30-01-2024	Card	ngosi@gmail.com	Lagos-Nig
9 iture		2	45000	90000	2024.02.01	Card	ibrahimbello@gmail.com	Kano-Nig
10 e Equipment		1	80000	80000	2024.02.25	Card	adawhite@gmail.com	Ibadan-Ni
11 ironics		2	350000	700000	12/01/2024	Card	johnndoe@gmail.com	Lagos-Nig
12 ironics		3	150000	450000	26/01/2024	Transfer	kevinhart@gmail.com	Abuja-Nig
13 ironics		2	95000	190000	27-01-2024	Cash	blissingsam@gmail.com	Lagos-Nig
14 iture		1	125000	125000	2024/01/28	Card	michaelking@gmail.com	Kaduna-N
15 ironics		4	158000	632000	29/01/2024	transfer	ritadaniel@gmail.com	Port Haro
16 ironics		2	120000	240000	30-01-2024	Card	gracepeter@gmail.com	Lagos-Nig
17 iture		7	27000	189000	31/01/2024	Cash	musabrahim@gmail.com	Abuja-Nig
18 ironics		2	150000	300000	2024-02-01	Transfer	janetcole@gmail.com	Kano-Nig
19 ironics		1	360000	360000	02/02/2024	Card	pauladams@gmail.com	Lagos-Nig
20 iture		3	50000	150000	2024/02/03	CASH	sandralee@gmail.com	Ibadan-Ni

14 COLUMNS, 20 ROWS Column profiling based on top 1000 rows

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Group Use First Row as Headers Count Rows Table

Transpose Reverse Rows Reverse Columns Count Rows Table

Data Type Date Detect Data Type Fill Pivot Column Convert to List Any Column

Replace Values Unpivot Columns Move Split Column Format Extract Parse Text Column

Merge Columns Statistics Standard Scientific Rounding Information Number Column

Trigonometry Date Time Duration Expand Aggregate Extract Values Date & Time Column Structured Column

Queries dirty_retail_sales...

Table.TransformColumnTypes(#Replaced Value9,{{"Order_Date",type date}},{"en-NG"})

Category 1.2 Quantity Unit_Price Total_Sales Order_Date Payment_Method Customer_Email

valid 100% Valid 100% Valid 100% Valid 1.2 Decimal Number valid 95% Valid 100% Valid
error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error
empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty

distinct, 0 unique 7 distinct, 3 unique 17 distinct, 14 unique 17 distinct, 14 unique 18 distinct, 16 unique 7 distinct, 3 unique 13 distinct, 18 unique 7 distinct

Category	1.2	Quantity	Unit_Price	Total_Sales	Order_Date	Payment_Method	Customer_Email	Customer_Location
1 ironics		2	350000	700000	1/12/2024	Card	johnndoe@gmail.com	Lagos-Nig
2 ironics		3	150000	450000	1/15/2024	Cash	maryann@yahoo.com	Abuja-Nig
3 ironics		1	400000	400000	1/15/2024	Transfer	peterobi@gmail.com	Kano-Nig
4 e Equipment		4	85000	340000	8/1/2024	card	chioma@gmail.com	Lagos-Nig
5 iture		2	120000	240000	1/20/2024	CASH	davidmusa@gmail.com	Abuja-Nig
6 ironics		5	160000	800000	1/21/2024	Transfer	estherali@gmail.com	Kaduna-N
7 ironics		1	355000	355000	1/22/2024	Card	samuelade@gmail.com	Port Haro
8 iture		6	25000	150000	1/23/2024	Card	ngosi@gmail.com	Lagos-Nig
9 iture		2	45000	90000	1/24/2024	Cash	ibrahimbello@gmail.com	Kano-Nig
10 e Equipment		1	80000	80000	1/25/2024	Card	adawhite@gmail.com	Ibadan-Ni
11 ironics		2	350000	700000	1/12/2024	Card	johnndoe@gmail.com	Lagos-Nig
12 ironics		3	155000	465000	1/12/2024	Card	johnndoe@gmail.com	Lagos-Nig
13 ironics		2	95000	190000	1/26/2024	Transfer	kevinhart@gmail.com	Abuja-Nig
14 iture		1	125000	125000	1/27/2024	Cash	blissingsam@gmail.com	Lagos-Nig
15 ironics		4	158000	632000	1/28/2024	Card	michaelking@gmail.com	Kaduna-N
16 ironics		2	120000	240000	1/29/2024	transfer	ritadaniel@gmail.com	Port Haro
17 iture		7	27000	189000	1/30/2024	Card	gracepeter@gmail.com	Lagos-Nig
18 ironics		2	150000	300000	1/31/2024	Card	musabrahim@gmail.com	Abuja-Nig
19 ironics		1	360000	360000	2/1/2024	Transfer	janetcole@gmail.com	Kano-Nig
20 iture		3	50000	150000	2/2/2024	Card	pauladams@gmail.com	Lagos-Nig
					2/3/2024	CASH	sandralee@gmail.com	Ibadan-Ni

14 COLUMNS, 20 ROWS Column profiling based on top 1000 rows

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Query Settings

PROPERTIES

Name dirty_retail_sales_dataset

APPLIED STEPS

Source

Promoted Headers

Changed Type

Removed Duplicates

Trimmed Text

Split Column by Position

Changed Type1

Removed Columns

Renamed Columns

Uppercased Text

Filtered Rows

Capitalized Each Word

Replaced Value

Replaced Value1

Filtered Rows1

Replaced Value2

Filtered Rows2

Capitalized Each Word1

Removed Columns1

Replaced Value3

Replaced Value4

Replaced Value5

Replaced Value6

Changed Type2

Replaced Value7

Replaced Value8

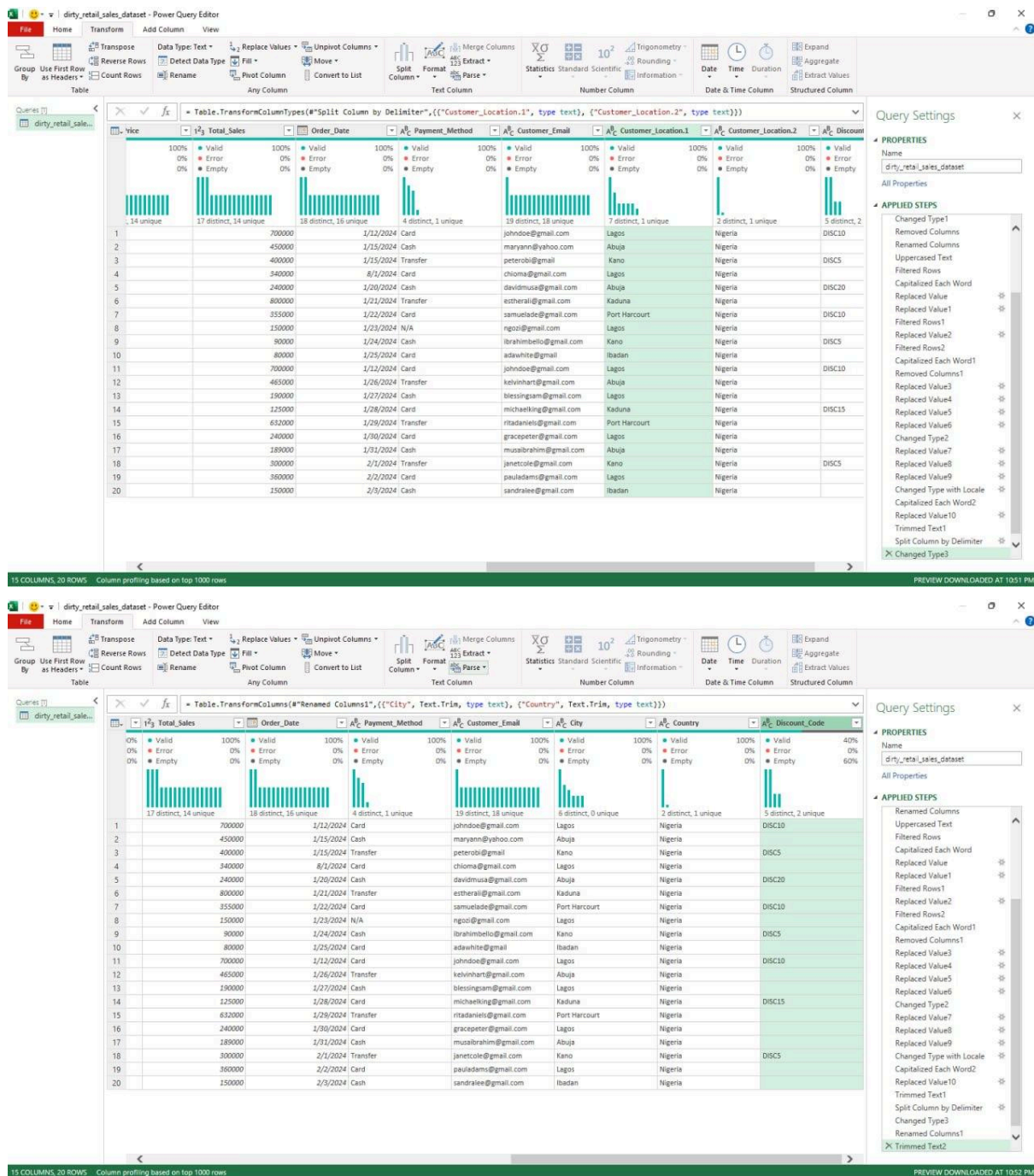
Replaced Value9

Changed Type with Locale

For column CUSTOMER LOCATION;

Problem; There were spaces before texts and in between texts and all of these inconsistencies.

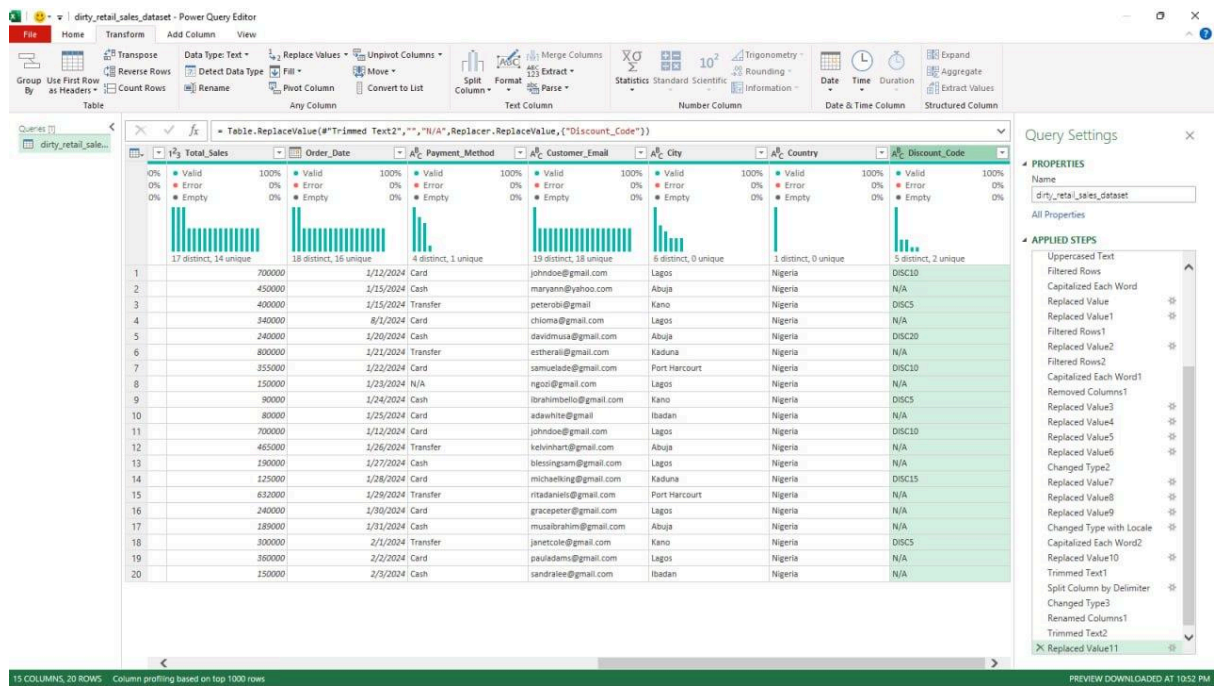
Solution; First, I splitted the column to two to make It State/City, and Country. Afterwards, I trimmed both columns to remove unwanted spaces.



For column DISCOUNT

Problems; There were blanks in some records.

Solution; The blank means they weren't given any discount. So I just replaced it with N/A.



ANALYSIS

The analysis was done after I cleaned the dataset and here are some key findings.

1) Phones and Chairs are the highest selling products.

	A	B	C
1	Row Labels	Sum of Quantity	
2	Phone	17	
3	Chair	13	
4	Tablet	7	
5	Laptop	7	
6	Printer	5	
7	Desk	3	
8	Monitor	2	
9	Grand Total	54	
10			

2) Gender (Male and Female) have the same distribution of orders.

12	Row Labels	Count of Order_ID
13	F	10
14	M	10
15	Grand Total	20
16		
17		
18	Row Labels	Sum of Total_Sales
19	F	3,500,000
20	M	3,500,000

3) Lagos and Abuja are the T2 selling states in terms of feve

17		
18	Row Labels	Sum of Total_Sales
19	Lagos	2,680,000
20	Abuja	1,344,000
21	Port Harcourt	987,000
22	Kaduna	925,000
23	Kano	790,000
24	Ibadan	230,000
25	Grand Total	6,956,000
26		